

Capital Markets

2025 .JITSTP

Basic Information

Course Title	Capital Markets
Instructor	Professor Alexei Chekhlov, Ph.D., Professor of Mathematics Department, Columbia University
Prerequisites	Basic statistics, basic calculus, and basic Excel knowledge
Required Text & Tools	Bodie, Kane, Marcus "Investments" (no need to purchase, I will distribute high quality pdf copies of the chapters I intend to cover) Authentic Microsoft Excel with authentic Add-Ins such as Microsoft Solver
Grading Criteria	Exam: 50% Homework: 40% Class Participation: 10%
Course Key Words	Asset Allocation, Risk Management, Asset Pricing, Index Model, Investment Strategy, Behavioral Finance, ESG Investing

Schedule

No.	Topics
Lecture 1	Asset classes and financial instruments
Lecture 2	How securities are traded?
Lecture 3	Real and nominal interest rates
Lecture 4	Capital allocation to risky assets
Lecture 5	Diversification and portfolio risk
Lecture 6	Some Markowitz procedure drawbacks
Lecture 7	Capital Asset Pricing Model
Lecture 8	Arbitrage Pricing Theory
Lecture 9	Efficient market hypothesis
Lecture 10	Behavioral finance and technical analysis